

Network Address Translation (Part 1)

- Topics covered:
 - Private IPv4 Addresses
 - Intro to NAT
 - Static NAT
 - Static NAT Configurations

Private IPv4 Addresses

Used of Private IPv4 Addresses

Previously, IPv6 was introduced as the long-term solution to IPv4 address exhaustion. However, migrating the entire Internet to IPv6 takes time, so IPv4 continues to be extended using several techniques.

The first solution is **CIDR (Classless Inter-Domain Routing)**, which removes the limitations of IPv4 address classes by allowing any prefix length.

The other two important solutions are:

1. **Private IPv4 addresses**
2. **NAT (Network Address Translation)**

RFC 1918 Private IPv4 Address Ranges

RFC (Request For Comments) documents define Internet standards.

RFC 1918 specifies these private IPv4 address ranges:

Address Range	Usable Addresses
10.0.0.0/8	10.0.0.0 – 10.255.255.255
172.16.0.0/12	172.16.0.0 – 172.31.255.255
192.168.0.0/16	192.168.0.0 – 192.168.255.255

These are simply address ranges. They can be divided into subnets in any way required using CIDR.

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Address classes are no longer important in modern networks because CIDR is now used.

Characteristics of Private IP Addresses

1. Can be freely used inside internal networks.
2. Do not need to be globally unique.
3. Commonly assigned to home and enterprise devices.
4. Cannot be routed over the Internet.
5. Internet Service Providers (ISPs) discard packets whose source or destination uses private IP addresses.

Example of a Private IP Configuration (CLI)

```
C:\Users\user>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.0.167
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
```

This example shows:

1. PC IP address: 192.168.0.167
2. Subnet mask: 255.255.255.0 (/24)
3. Default gateway (router): 192.168.0.1

Why NAT Is Needed

Private IP addresses cannot travel across the Internet.

If two different networks both use the same private subnet (for example, **192.168.0.0/24**), duplicate addresses would exist on the Internet, making communication impossible.

Without NAT



Problems:

1. Duplicate private IP addresses exist.
2. Private IP addresses cannot be routed on the Internet.

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Public IP Addresses Solve the Problem

Routers use unique public IP addresses.



Key points:

1. Private addresses may be duplicated.
2. Public addresses **must** be globally unique.
3. Internal devices communicate using the router's public address after NAT.



Network Address Translation

How Network Address Translation Work?

NAT is used to modifies the:

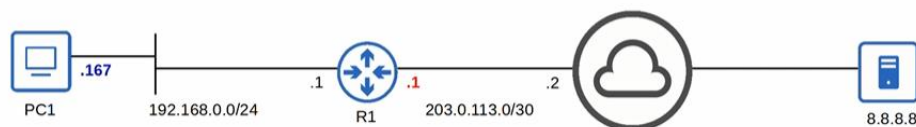
1. Source IP address
2. Destination IP address

depending on the type of NAT being used.

The primary purpose of NAT is, allow hosts using private IP addresses to communicate over the Internet.

For the CCNA you have to understand source NAT and how to configure it on Cisco routers.

Source NAT Example



PC1 wants to communicate with 8.8.8.8.

Step 1: Packet created by PC1:

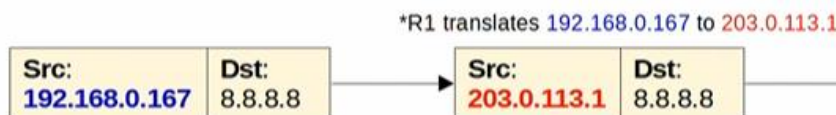
Src:	Dst:
192.168.0.167	8.8.8.8

PC1 sends the packet to its default gateway (R1).

Step 2: R1 performs Source NAT.

Before NAT

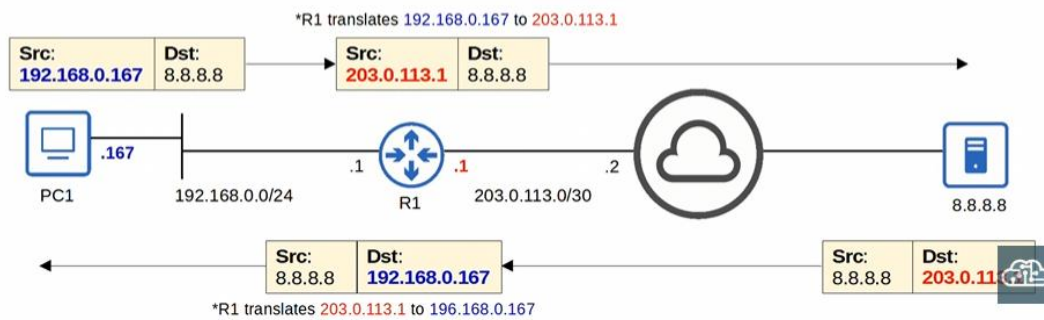
After NAT



R1 changes only the source IP address, The destination remains unchanged.

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Step 3: Packet travel across internet and reach 8.8.8.8

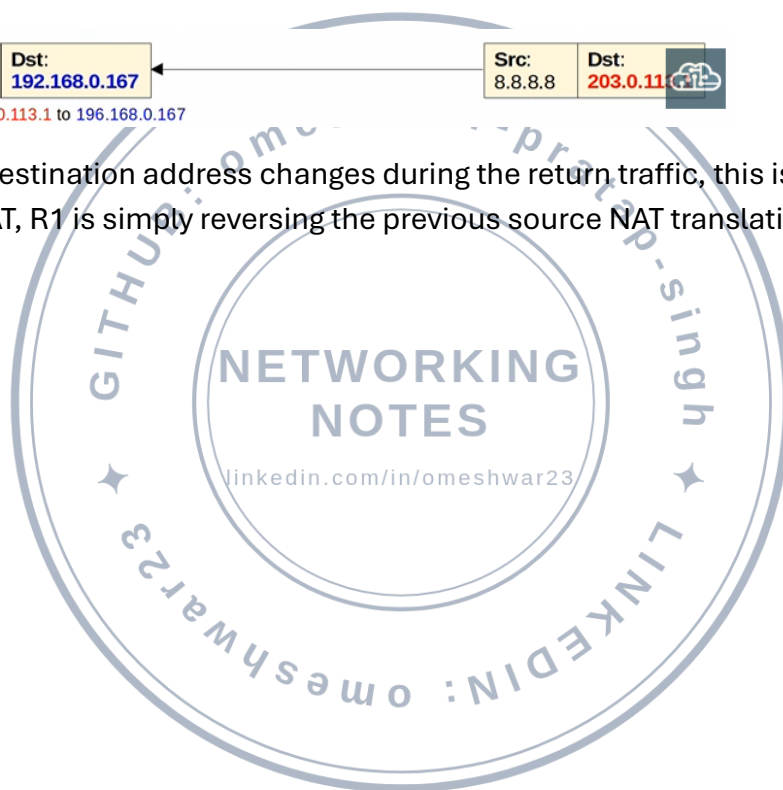


Step 4: After 8.8.8.8 response back to the IP 192.168.0.167

R1 reverses the translation.



Although the destination address changes during the return traffic, this is not destination NAT, R1 is simply reversing the previous source NAT translation.



Static NAT

What is Static NAT?

Static NAT creates a **permanent one-to-one mapping** between two IP addresses.

Typically: One Private IP $\updownarrow$ One Public IP

Although any address can be translated to another address, it is commonly used to map: Private IP $\rightarrow$ Public IP

Inside Local and Inside Global Addresses

Two important NAT terms:

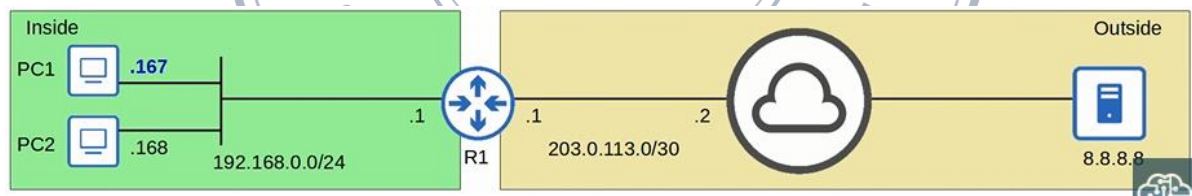
Inside Local: The IP address of the inside host as seen inside the local network.

It is: Actually, configured on the host, usually a private IP address.

Inside Global: The IP address of the inside host as seen by outside hosts.

It is: The address after NAT, usually a public IP address.

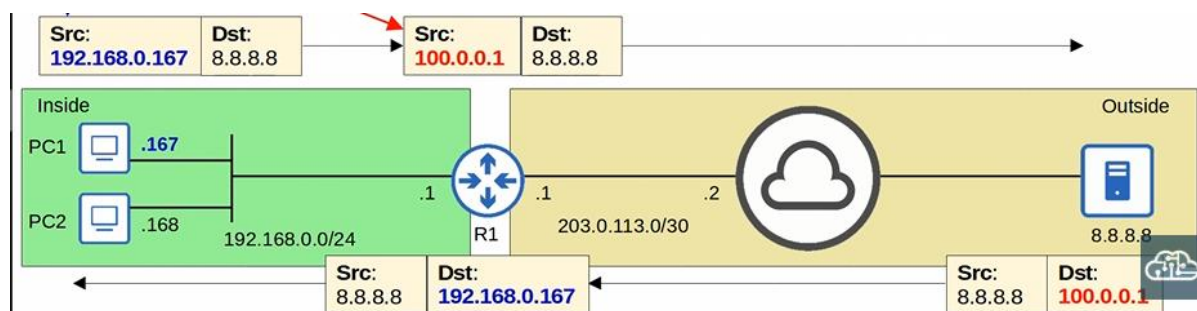
Static NAT Example



PC 1 wants to communicate with Server 8.8.8.8

Before NAT

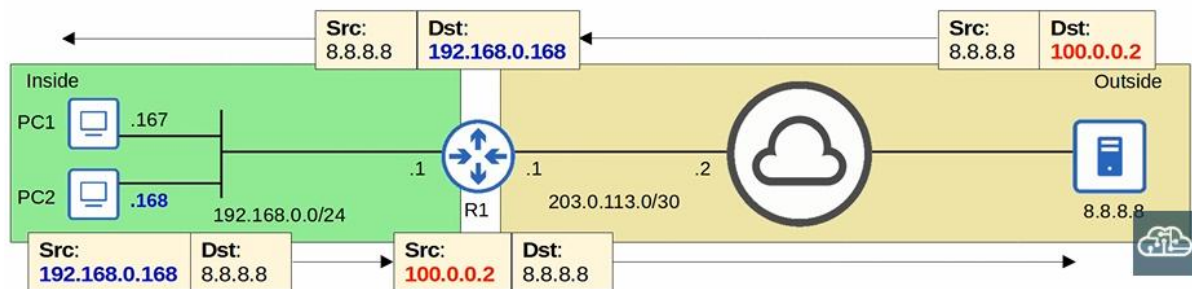
After NAT



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Multiple Hosts with Static NAT

Suppose another host (PC2) also needs Internet access.



Mappings

(192.168.0.167 ---> 100.0.0.1) PC1

(192.168.0.168 ---> 100.0.0.2) PC2

For PC2:

1. Inside Local: 192.168.0.168
2. Inside Global: 100.0.0.2

Each inside device requires its own dedicated public IP address.

Limitation of Static NAT

Static NAT allows private devices to communicate over the Internet.

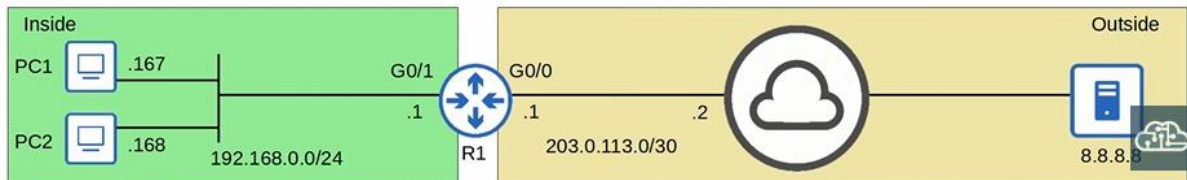
However:

1. It requires a **one-to-one mapping**.
2. Every internal device needs its own public IP address.
3. Therefore, Static NAT does **not** conserve public IPv4 addresses effectively compared to other NAT methods.

Configuring Static NAT

How To Configure Static NAT

Static NAT requires only a few configuration commands.



1. Configure the Inside Interface

Use the **ip nat inside** command in interface configuration mode to identify the interface connected to the internal network.

In this network:

1. G0/1 is connected to the internal network.
2. Therefore, G0/1 is configured as the inside interface.

```
R1(config)#int g0/1
R1(config-if)#ip nat inside
```

Define the 'inside' interface(s) connected to the internal network.

2. Configure the Outside Interface

Use the **ip nat outside** command on the interface connected to the external network.

In this network:

1. G0/0 is connected to the external network.
2. Therefore, G0/0 is configured as the outside interface.

```
R1(config-if)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#exit
```

Define the 'outside' interface(s) connected to the external network.

3. Configure Static One-to-One Mappings

```
R1(config)#ip nat inside source static 192.168.0.167 100.0.0.1
R1(config)#ip nat inside source static 192.168.0.168 100.0.0.2
R1(config)#exit
```

Configure the one-to-one IP address mappings.
ip nat inside source static inside-local-ip inside-global-ip

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This creates the following mappings:

Inside Local Inside Global

192.168.0.167 100.0.0.1

192.168.0.168 100.0.0.2

As a result:

1. PC1's IP address is translated to 100.0.0.1.
2. PC2's IP address is translated to 100.0.0.2.

Note: Public IP addresses must belong to you or your organization. Unlike private addresses, they cannot be chosen freely.

Network After Configuration

4. Verifying NAT Translations

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp 100.0.0.1:56310    192.168.0.167:56310 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167    ---               ---
udp 100.0.0.2:62321    192.168.0.168:62321 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168    ---               ---
```

Explanation

Static Entries: When Static NAT is configured, the one-to-one mappings are always present in the translation table.

Notice that Cisco displays:

- **Inside Global**
- **Inside Local**

instead of showing Inside Local first.

The permanent mappings are:

Inside Global Inside Local

100.0.0.1 192.168.0.167

100.0.0.2 192.168.0.168

These entries remain in the table permanently.

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Dynamic Translation Entries: Whenever NAT actually translates packets, additional dynamic entries appear.

In this example:

1. PC1 sent traffic to **8.8.8.8**.
2. PC2 sent traffic to **8.8.8.8**.

Therefore, dynamic entries were created.

Translation Table Columns

1. Pro: Shows the protocol used.

2. Inside Global: Shows the translated IP address of the inside host.

Example: 100.0.0.1:56310

The number after the colon is the **port number**.

For Static NAT, the port numbers are not very important.

3. Inside Local: Shows the actual IP address configured on the inside host.

Example: 192.168.0.167:56310

Although the IP address is translated, Static NAT does **not** translate the port numbers.

The port numbers remain unchanged.

4. Outside Local: The IP address of the outside host from the perspective of the local (inside) network.

For PC1: Outside Local = 8.8.8.8

5. Outside Global: The IP address of the outside host from the perspective of the outside network.

For this server:

Outside Global = 8.8.8.8

Unless **Destination NAT** is used:

Outside Local = Outside Global

For the CCNA, these two addresses can be expected to be the same.

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Understanding the Four NAT Address Types

The four NAT terms describe two different concepts.

Host Location

1. **Inside** → Host is on the internal network.
2. **Outside** → Host is on the external network.

In this example:

- PC1 and PC2 are inside hosts.
- The server at 8.8.8.8 is an outside host.

Perspective

- **Local** → Viewed from the local (inside) network.
- **Global** → Viewed from the outside network.

Example:

Address	Meaning
192.168.0.167	PC1's address from the local network (Inside Local)
100.0.0.1	PC1's address from the outside network after NAT (Inside Global)

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp 100.0.0.1:56310    192.168.0.167:56310 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167      ---                ---
udp 100.0.0.2:62321    192.168.0.168:62321 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168      ---                ---
```

Unless **destination NAT** is used, these two addresses will be the same.

→ **Inside Local** = The IP address of the *inside* host, from the perspective of the local network
*the IP address actually configured on the inside host, usually a private address

→ **Inside Global** = The IP address of the *inside* host, from the perspective of *outside* hosts
*the IP address of the inside host after NAT, usually a public address

→ **Outside Local** = The IP address of the *outside* host, from the perspective of the local network

→ **Outside Global** = The IP address of the *outside* host, from the perspective of the outside network

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Clearing Dynamic NAT Entries

```
R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
udp 100.0.0.1:56310    192.168.0.167:56310 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.1          192.168.0.167    ---              ---
udp 100.0.0.2:62321    192.168.0.168:62321 8.8.8.8:53         8.8.8.8:53
--- 100.0.0.2          192.168.0.168    ---              ---

R1#clear ip nat translation *

R1#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 100.0.0.1          192.168.0.167    ---              ---
--- 100.0.0.2          192.168.0.168    ---              ---
```

Explanation

The command: `clear ip nat translation *`, removes all **dynamic** NAT translations.

Although only Static NAT was configured, dynamic entries are temporarily created whenever the static mappings are actively used.

Normally, these dynamic entries are automatically removed after timing out when communication stops, Using this command removes them immediately.

The permanent Static NAT entries remain, They do not time out and cannot be removed unless the corresponding: `ip nat inside source static` configuration commands are deleted.

Viewing NAT Statistics

```
R1#show ip nat statistics
Total active translations: 2 (2 static, 0 dynamic; 0 extended)
Peak translations: 4, occurred 02:29:00 ago
Outside interfaces:
  GigabitEthernet0/0
Inside interfaces:
  GigabitEthernet0/1
Hits: 34 Misses: 0
CEF Translated packets: 30, CEF Punted packets: 4
Expired translations: 4
Dynamic mappings:

Total doors: 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
```

Explanation

Total active translations: 2

1. Static: 2
2. Dynamic: 0
3. Extended: 0

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Since the dynamic entries were cleared, only the two static mappings remain.

Peak translations: 4, The highest number of translations that existed in the NAT table.

Outside Interface: GigabitEthernet0/0

Inside Interface: GigabitEthernet0/1

Static NAT Commands Summary

```
R1(config-if)# ip nat inside
R1(config-if)# ip nat outside
R1(config)# ip nat inside source static inside-local-ip inside-global-ip
R1# show ip nat translations
R1# show ip nat statistics
R1# clear ip nat translation *
```

